

GhostNet-AI — Alternative Solution Architecture

Different Problem-Solving Approaches for SIH26057

Yes. Looking at **your exact SIH26057 problem statement** and your current GhostNet-AI architecture, there are several ways to solve the same problem that are **different from a standard “YOLO on sonar image + dashboard” approach**.

The official requirement is broader than a particular model: the system needs to ingest SSS data, separate man-made objects from natural seabed, handle sonar noise/shadows/dropouts, provide confidence, geolocate detections, and expose the results through a usable interface. ([SIH 2026 Problem Statements](https://sih2026.vuce.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 · AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH 2026 Problem Statements"))

Also, current public SIH26057 work already includes approaches centered on YOLO/object detection, calibrated confidence, geotagging, dashboards and edge CPU inference, so merely adding another YOLO model is not much of a differentiator. ([GitHub](https://github.com/Rehan9599/Sonar-DrishTi?utm_source=chatgpt.com "GitHub - Rehan9599/Sonar-DrishTi · GitHub")) Recent GhostVision research similarly combines low-cost SSS, object detection and mapping. ([MDPI](https://www.mdpi.com/2077-1312/14/10/951?utm_source=chatgpt.com "GhostVision: Democratizing Derelict Gear Detection Using Low-Cost Sonar and Artificial Intelligence | MDPI"))

So I would think about **changing the way the problem is solved**, not just changing the model.

1. Instead of “object detection,” build a SONAR ANOMALY ENGINE

This is probably the biggest conceptual change.

Instead of asking:

“Can YOLO detect a ghost net?”

ask:

Is this sonar region inconsistent with the surrounding natural seabed?

Architecture:

```

RAW SSS
↓
Sonar normalization
↓
Natural seabed representation
↓
Anomaly map
↓
Candidate regions
↓
Man-made / natural classification

```

The system first learns what **normal seabed** looks like.

Then:

Normal seabed
■■■■■■■■■■■■■■■■■■■■

Suspicious region
■■■■■■■■■■■■■■■■■■■■

Why this is interesting

Ghost nets are difficult because exact labelled examples are limited. The SIH statement itself emphasizes the problem of distinguishing artificial objects from rock formations, sand ripples and ridges. ([SIH 2026 Problem Statements](https://sih2026.vuce.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 · AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH 2026 Problem Statements"))

So you can say:

***"We don't require every possible debris type to be explicitly known beforehand; we first detect acoustic anomalies and then determine whether they appear anthropogenic."**

That gives you an **open-world** approach.

2. Self-supervised / few-shot SONAR learning

A different way to attack the data problem:

Instead of needing thousands of labelled ghost-net images, use large amounts of **unlabelled sonar**.

```
UNLABELLED SONAR
↓
Self-supervised learning
↓
Sonar feature encoder
↓
Small labelled dataset
↓
Fine-tune classifier/detector
```

The practical idea is:

learn sonar patterns first → teach the model specific objects second.

For your project, this can be especially useful because public SSS datasets are often much richer in generic seabed/shipwreck/aircraft imagery than in actual ghost nets. Current project discussions and datasets reflect this imbalance. ([SIH Buddy](https://www.sihbuddy.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 — AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH Buddy"))

Your pitch

***"Our model learns the language of sonar before learning the vocabulary of debris."**

That is a much better story than "we trained YOLO."

3. Physics + AI hybrid model

This is another very strong direction.

Don't ask AI to learn everything from pixels.

Give it **explicit sonar evidence**.

For each candidate calculate:

```
Object shape
+
Acoustic shadow
+
Shadow length
+
Intensity profile
+
Local seabed texture
+
Target height estimate
+
Track persistence
```

Then:

```
AI DETECTOR
+
SONAR FEATURES
+
PHYSICS RULES
↓
FINAL SCORE
```

Example:

```
YOLO = 84%
Shadow consistency = High
Target persistence = High
Seabed context = Artificial
Sonar quality = Good

→ HIGH EVIDENCE
```

This is much stronger than claiming “physics-aware YOLO.” There is already research using acoustic shadow information in SSS object detection, so your novelty should be the **integrated evidence-fusion workflow**, not the existence of shadow features itself. ([MDPI](https://www.mdpi.com/2077-1312/14/10/951?utm_source=chatgpt.com "GhostVision: Democratizing Derelict Gear Detection Using Low-Cost Sonar and Artificial Intelligence | MDPI"))

4. Multi-frame thinking — treat the sonar survey as VIDEO

This is one of my favorite ideas for your project.

Most teams will think:

```
Image → AI → Detection
```

You can think:

```
PING 1
PING 2
PING 3
PING 4
PING 5
```

↓
SAME TARGET?

Then track the object through the sonar stream.

```
Frame 101 ■■■■  
Frame 102 ■■■■  
Frame 103 ■■■■ → TARGET #21  
Frame 104 ■■■■  
Frame 105 ■■■■
```

Result

Instead of:

5 detections

you report:

1 persistent target observed across 5 pings

This can reduce duplicates and give you stronger evidence.

5. Track-to-map fusion

Take the previous idea one step further.

Each detection gives a noisy location:

```
Frame 1 → (x1, y1)  
Frame 2 → (x2, y2)  
Frame 3 → (x3, y3)  
Frame 4 → (x4, y4)
```

Fuse them:

```
      x  
    x  x  
      ●  
    x  x  
  FINAL  
LOCATION
```

Then output:

Estimated position: X,Y

Uncertainty: ±R meters

This turns your geolocation module into an actual **estimation problem** rather than simply extracting a coordinate from metadata.

That is particularly relevant because the SIH statement asks for geolocation and actionable localized information. ([SIH 2026 Problem Statements](https://sih2026.vuce.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 · AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH 2026 Problem Statements"))

6. “Three opinions” AI

Instead of trusting one model:

```
YOLO
+
Anomaly Detector
+
Acoustic Evidence Model
↓
Agreement?
```

Example:

Detector	91%
Anomaly model	84%
Shadow model	93%

→ Strong agreement

But:

Detector	93%
Anomaly model	51%
Shadow model	28%

→ Conflict

→ HUMAN REVIEW

This leads naturally to your:

CONFIRM / REVIEW / ABSTAIN

idea.

That is much more defensible than pretending model confidence alone represents certainty.

7. Retrieval-based “Similar Sonar” search

This is a very different user experience.

When the operator clicks a target:

```
TARGET #27
Possible Ghost Net
```

the system searches a sonar embedding database:

```
SIMILAR TARGETS

91% similarity → known ghost gear
87% similarity → rope/net structure
83% similarity → natural seabed
```

So the AI says:

"This target resembles these previously reviewed examples."

This is useful for human review because the operator gets **evidence from previous cases**, not just a prediction.

8. Unknown-object discovery

This is a very strong answer to the fact that real oceans contain objects you didn't train for.

Classes:

```
KNOWN
■■■ Ghost Net
■■■ Wreck
■■■ Pipe
■■■ Cylinder
■■■ Aircraft

UNKNOWN
■■■ Possible Man-Made Anomaly
```

Then:

```
Unknown target
↓
Human labels it
↓
New class/example
↓
Future training
```

This also connects naturally to your human-in-the-loop concept.

9. Active-learning system

Instead of collecting random more data, let the AI tell you:

***"These are the 100 examples I most need a human to label."**

For example:

```
Model uncertainty
■■■■■■■■■■ Target 17
■■■■■■■■■■ Target 41
■■■■■■■■■■ Target 83
■■■■■■■■■■ Target 96
```

Operator labels those.

Then retrain.

```
AI
↓
Uncertain examples
↓
Human annotation
↓
```

```

graph TD
    A[Hard-example dataset] --> B[Retrain]
    B --> C[Better AI]

```

This makes your system **improve from field operations.**

10. Sonar Quality Intelligence

Before AI detection, assess whether the sonar itself is usable.

SONAR QUALITY

■■■■■■■■■■■■■■■■■■■■■

Noise	32%
Contrast	81%
Dropout	14%
Shadow clarity	76%
Navigation	94%

Overall:
GOOD

Then your pipeline changes:

```
GOOD
→ normal inference

MEDIUM
→ conservative inference

POOR
→ warning + review
```

This is a genuinely useful safety layer because the PS explicitly calls out speckle, varying resolution and motion-related dropouts. ([SIH 2026 Problem Statements](https://sih2026.vuce.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 · AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH 2026 Problem Statements"))

11. “Survey Coverage Intelligence”

This isn't just about detections.

Your map could answer:

Where did we search?

```
GREEN  = surveyed
GREY   = not surveyed
YELLOW = poor-quality survey
RED    = suspicious target
```

Now the operator sees:

Detection map + survey coverage map

This helps avoid the misleading assumption:

“No detection = no debris.”

Instead:

“No detection in a high-quality surveyed region.”

That’s a much more mature interpretation.

12. Repeat-survey change detection

This is another completely different solution direction.

Survey an area in June:

Survey A
→ Target map A

Survey it again in September:

Survey B
→ Target map B

Then:

A vs B
↓
NEW TARGETS
REMOVED TARGETS
PERSISTENT TARGETS
CHANGED AREAS

Recent SSS research is actively studying change detection across repeat tracks, so this is technically credible as an extension. ([DOI](https://doi.org/10.15587/1729-4061.2025.346940?utm_source=chatgpt.com "Change detection in side-scan sonar imagery based on deep learning feature matching methods | Eastern-European Journal of Enterprise Technologies"))

This turns your project from:

“find debris once”

into:

“monitor the seabed over time.”

13. Recovery planning instead of only reporting

Current PS asks for actionable localized data and reports. ([SIH 2026 Problem Statements](https://sih2026.vuce.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 · AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH 2026 Problem

Statements"))

You can take the next step:

```
19 detected targets
↓
12 confirmed
↓
Priority calculation
↓
Recovery mission planning
```

Output:

```
MISSION 01
START
↓
Target 08
↓
Target 14
↓
Target 21
↓
Target 27
↓
BASE
```

So your system becomes:

Detection → Decision → Mission planning

rather than:

Detection → CSV

14. Natural-language SonarOps Agent

Your idea from the previous message fits here.

But I would use the agent as the **orchestrator**, not the actual detector.

Operator:

"Process survey_014, identify likely ghost nets, ignore low-confidence results, map confirmed targets and create a report."

Agent:

```
PLAN
↓
Validate XTF
↓
Assess sonar quality
↓
Preprocess
↓
Run anomaly detection
↓
Run object detector
↓
Fuse evidence
```

```

↓
Track targets
↓
Geolocate
↓
Request review for uncertain cases
↓
Generate report

```

That is much stronger than an agent simply clicking your website.

...

15. Agent that explains failures

This would be a very good demo.

Suppose:

GPS metadata missing

Instead of crashing:

"GPS data is unavailable. Object detection can continue, but geolocation will be marked unavailable. Continue?"

Or:

Poor sonar quality

Agent:

This segment contains significant signal dropout. I can process it using conservative detection thresholds and send uncertain candidates for review.

That combines your **agent + uncertainty + graceful degradation** into one feature.

— — —

16. Build a digital “Target Passport”

Every confirmed target gets an automatically generated record:

[illegible]

This would look excellent in both your software and PPT.

17. A particularly clever approach: “Natural-first” classification

Most systems ask:

“Is this ghost net?”

You can reverse it:

Step 1

Determine whether the region is explainable by natural seabed structure.

Natural?
YES → discard
NO → candidate

Step 2

Only suspicious regions are passed to the object classifier.

SEABED
↓
Naturalness model
↓
Suspicious region
↓
Object classification

This can reduce false positives and compute requirements.

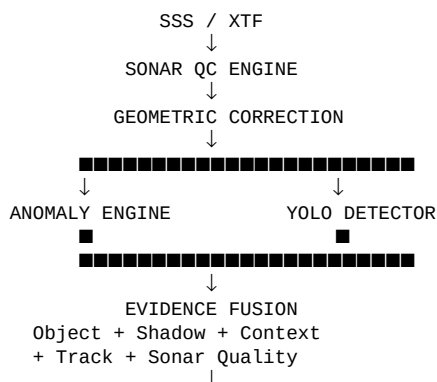
Conceptually:

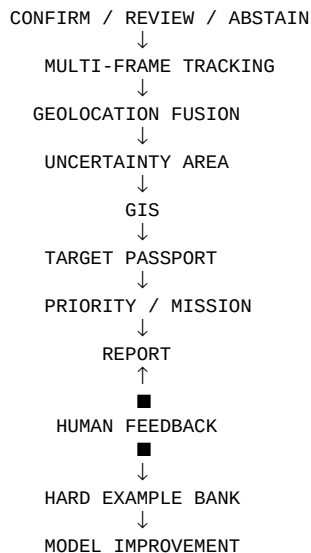
****Reject the obvious natural world first; spend expensive AI on the unexplained remainder.****

That’s an interesting system architecture.

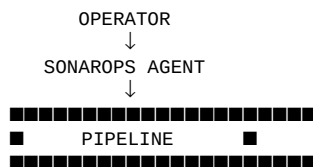
18. My recommended final architecture for YOUR team

I would combine the strongest ideas rather than trying to build 15 independent features:





And put the **SonarOps Agent** above the whole pipeline:



If you want your solution to look fundamentally different

I would frame GhostNet-AI as **three layers**, rather than as one AI model:

Layer 1 — SEE

Sonar Intelligence

Anomaly detection + object detection + sonar physics.

Layer 2 — TRUST

Evidence Intelligence

Confidence + evidence fusion + uncertainty + human review.

Layer 3 — ACT

Mission Intelligence

Geolocation + target passport + prioritization + agent + reporting.

So your one-line architecture becomes:

****SEE → TRUST → ACT****

That's a stronger product concept than:

****IMAGE → YOLO → BOX****

And it still satisfies the actual SIH requirement: automated SSS analysis, artificial-vs-natural discrimination, confidence/noise filtering, localization, reporting and dashboard-based visualization. ([SIH 2026 Problem Statements](https://sih2026.vuce.in/ps/SIH26057?utm_source=chatgpt.com "SIH26057 · AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery | SIH 2026 Problem Statements"))

What I would actually build

For a hackathon, I'd make these **five features genuinely work end-to-end**:

- 1. Sonar Anomaly Engine**
- 2. YOLO + Acoustic Evidence Fusion**
- 3. Multi-frame Target Tracking + Geolocation Fusion**
- 4. Confirm / Review / Abstain with Target Passport**
- 5. SonarOps Agent that orchestrates the entire workflow**

Everything else can be shown as an extension.

That combination gives you a **different problem-solving philosophy**, not just a different UI.